Suk Yee Yong

Artificial intelligence & Machine learning specialist ⋆ Astrophysicist ⋆ Data scientist

Scientific Computing
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Profile Summary

Data scientist with 10+ years of experience and a PhD in Astrophysics, specialising in the application of artificial intelligence, statistical modelling, and advanced analytics to complex scientific and business challenges. From the macro scale of the universe to the molecular level of DNA, experience spans astronomy, healthcare, environmental science, and climate analytics. Brings expertise in neural networks, anomaly detection, uncertainty quantification, large language models, and cross-domain AI development, translating rigorous scientific research into practical, data-driven solutions with real-world impact.

Research Interests

⋆ anomaly detection ⋆ large language model ⋆ machine learning ⋆ supervised learning
⋆ uncertainty quantification ⋆ unsupervised learning
⋆ black hole ⋆ quasars ⋆ data analysis ⋆ data visualisation

Work Experience

11|2025–present
Senior Data Science Specialist in Artificial Intelligence & Machine Learning
Scientific Computing, Commonwealth Scientific and Industrial Research Organisation (CSIRO), Brisbane, Australia

06|2023–11|2025
Artificial Intelligence & Machine Learning Specialist
Scientific Computing, CSIRO, Sydney, Australia

09|2022–06|2023
ARC Centre of Excellence for All Sky Astrophysics in 3 Dimensions (ASTRO 3D) Postdoctoral Fellow in Data Intensive Astronomy
Joint appointment between the School of Mathematics and Physical Science, Macquarie University, and Australian Astronomical Optics (AAO) Data Central, Sydney, Australia
- **Project:** Uncertainty Quantification of Virial Black Hole Mass
- **Project:** Retrieving Galaxy Kinematics from Spectroscopic Data
- **Project:** 3D Galaxy Shape Recovery using Neural Networks

09|2020–08|2022
CERC Postdoctoral Fellow in Machine Learning and Artificial Intelligence (MLAI) Future Science Platform (FSP)
MLAI FSP Object Detection Activity, Space and Astronomy, CSIRO, Sydney, Australia
- **Project:** Finding the unknowns in radio astronomy
- **Project:** Neural Network Kernel Design

03|2020–08|2020
Postdoctoral Research Fellow in Astrophysics
School of Physics, University of Melbourne, Melbourne, Australia
- **Project:** Quasar disk-wind phenomenology

10|2019–03|2020
Research Data Engineer
Centre for Eye Research Australia Limited, Royal Victorian Eye and Ear Hospital, Melbourne, Australia
- **Project:** The eyes and the sky — medical image analysis with astrophysics supercomputing

07|2019–09|2019
Astrophysics Data Science Intern
Astronomy and Data Computing Services (ADACS), Swinburne University of Technology, Melbourne, Australia

Project: Recover the 3D shape of galaxy using deep learning

07|2015–02|2016 **Research Assistant**
School of Physics, University of Melbourne, Melbourne, Australia

Project: Modelling the Biases in Virial Black Hole Mass Estimation

08|2012–11|2012 **Research Assistant**
Department of Physics, The Pennsylvania State University, University Park, USA

05|2012–08|2012 **Research Experience for Undergraduate (REU) Internship**
Department of Physics, The Pennsylvania State University, University Park, USA

05|2011–08|2011 **Research Assistant**
Department of Astronomy and Astrophysics, The Pennsylvania State University, University Park, USA

Education

11|2019–03|2020 **Specialist Certificate in Research Practice for Scientists**
University of Melbourne, Melbourne, Australia

03|2016–02|2020 **PhD in Physics (Astrophysics)**
University of Melbourne, Melbourne, Australia
Supervisor: Prof. Rachel Webster; Co-supervisor: Dr. Anthea King
Thesis: Nature of Quasar Disk-wind

07|2013–07|2015 **Master of Science (Physics) with Distinction**
University of Melbourne, Melbourne, Australia
Supervisor: Prof. Rachel Webster
Weighted average mark (WAM): 82.125/100.0
Thesis: Quasar Disk Wind Models and Emission Line Profiles

07|2010–12|2012 **Bachelor of Science with Distinction (Major in Physics, Minor in Mathematics)**
The Pennsylvania State University, University Park, USA
Grade point average (GPA): 3.85/4.0

Selected Publications

Total publications: 20; First author: 11

2026 Spillias, S., Rogers, J. G. D., Boschetti, F., Fulton, E. A., Guglielmo, M. **Yong, S. Y.**, et al. 2026. "Data-driven discovery of mechanistic ecosystem models with LLMs". *Methods in Ecology and Evolution* 17.4, pp. 1303–1321 [DOI]

2025 **Yong, S. Y.** and Takahashi, M. 2025. *FAIRe-fier: FAIR eDNA metadata verifier* [Service | App | Code]

2024 Budd, A. M., **Yong, S. Y.**, Heydenrych, M. J., Mayne, B., Berry, O., et al. 2024. "Universal prediction of vertebrate species age at maturity". *Communications Biology* 7.1, p. 1414 [DOI | Code]

2024 **Yong, S. Y.**, O'Grady, J., Gregory, R., and Lynton, D. 2024. "Regional-Scale Image Segmentation of Sandy Beaches in Southeastern Australia". *Remote Sensing* 16.18 [DOI | Data | Code]

2024 **Yong, S. Y.**, Harborne, K. E., Foster, C., Bassett, R., Poole, G. B., et al. 2024. "Galaxy 3D shape recovery using mixture density network". PASA 41, e033, e033 [DOI | Preprint]

2023 **Yong, S. Y.** and Ong, C. S. 2023. "Uncertainty quantification of the virial black hole mass with conformal prediction". MNRAS 524.2, pp. 3116–3129 [DOI | Preprint | Code]

2022 **Yong, S. Y.**, Hobbs, G., Huynh, M. T., Rolland, V., Petersson, L., et al. 2022. "SPARKESX: Single-dish PARKES data sets for finding the uneXpected - a data challenge". MNRAS 516.4, pp. 5832–5848 [DOI | Preprint]

2022 Tsuchida, R., **Yong, S. Y.**, Armin, M. A., Petersson, L., and Ong, C. S. 2022. "Declarative nets that are equilibrium models". In: *International Conference on Learning Representations* [Publisher]

2020 **Yong, S. Y.**, Webster, R. L., King, A. L., Bate, N. F., Labrie, K., et al. 2020. "Determining quasar orientation". MNRAS 491.1, pp. 1320–1334 [DOI | Preprint]

2018 **Yong, S. Y.**, King, A. L., Webster, R. L., Bate, N. F., O'Dowd, M. J., et al. 2018. "Using the Properties of Broad Absorption Line Quasars to Illuminate Quasar Structure". MNRAS 479.3, pp. 4153–4171 [DOI | Preprint]

Selected Awards and Achievements

2025 **Health and Biosecurity (H&B) Awards** in Outstanding Collaboration, *CSIRO H&B*

2024 **Non cash reward** for the technical expertise and contributions in exploring the intersections of generative AI and climate disclosure regulations, *CSIRO Responsible Innovation FSP*

2024 **Non cash reward** for the inspired and hard work in IMT Research Collaborations, demonstrating skills, problem-solving, and collaborative spirit that have significantly advanced the science, *CSIRO Environomics FSP*

2021 **Non cash reward** for demonstrating diverse, inclusive, belonging behaviour, *CSIRO MLAI FSP*

2021 **3rd place** in Pitch initiative for Machine Learning and Artificial Intelligence Reimagining Science (MARS) 2021, *CSIRO MLAI FSP*

2016–2020 **Melbourne Research Scholarship** Doctor of Philosophy, *University of Melbourne*

2019–2020 **Australian Postgraduate Research Intern**, *Cylite*

2019 **Astronomy and Data Computing Services Internship Program**, *Astronomy Australia Limited*

2019 **Best paper award** in track Astronomy and Astrophysics, Interdisciplinary Space Science and Others, *6th International Conference on Space Science and Communication*

2015 **Ramm Prize in Experimental Physics**, *University of Melbourne*

2011–2012 **Bert Elsbach Scholarship in Physics**, *Pennsylvania State University*

2010–2012 **Dean's List** for every semester, *Pennsylvania State University*

Selected Recent Seminars, Colloquia, and Invited Talks

Total presentations: 17

05|2026 CSIRO Applied AI Practice and Projects, *Machine Learning Community of Practice*, Virtual
- ⋆ **Talk:** AI xDomain Collaboration in Coastal Mapping and Biomedical Discovery

04|2025 Humans & Machines mini-Showcase, *CSIRO Data61 Humans and Machines*, Virtual
- ⋆ **Talk:** EcoActionAI — a tool to support climate risk and response reporting

07|2024 Language Technology Team Meeting, *CSIRO Data61 Language Technology*, Virtual
- ⋆ **Talk:** Maximising Potential — Embracing CSIRO High Performance Computing

01|2023 Global Malaysian Astronomers Convention 2023, *National Science Center*, Kuala Lumpur, Malaysia
- ⋆ **Invited talk:** Enabling Data-driven Astronomy - from data archive to data mining

12|2022 Science Expresso, *Science Cafe Kuala Lumpur*, Virtual
- ⋆ **Invited talk:** Can Artificial Intelligence Discover Extraterrestrial Intelligence?

06|2022 Object Detection Forum, *CSIRO MLAI FSP*, Virtual
- ⋆ **Talk:** Transformers with HuggingFace

03|2022 Science Seminars, *CSIRO MLAI FSP*, Virtual
- ⋆ **Talk:** Anomaly Detection for Time-domain Discoveries

09|2021 Machine Learning in Astronomy, *Western Sydney University*, Virtual
- ⋆ **Talk:** 3D Galaxy Shape using Mixture Density Network

Selected Recent Conference Contributions

Total presentations: 23

12|2025 Australasian Joint Conference on Artificial Intelligence 2025, *Workshop on AIML in Computational and Experimental Biology*, Canberra, Australia
- ⋆ **Paper and poster:** SciAgent OLab: Open Multi-Agent LLMs for Accelerating Biomedical Discovery
- ⋆ **Paper and oral, 2nd author:** Evolutionary Exploration with LLM-Generated Influenza Variants

05|2024 Computational Trends and Digital Science, Melbourne, Australia
- ⋆ **Talk:** Application of Artificial Intelligence and Uncertainty Quantification across Science Domains

12|2022 36th Neural Information Processing Systems 2022, *Workshop on Machine Learning and the Physical Sciences*, Virtual
- ⋆ **Paper and poster, co-author:** Detecting structured signals in radio telescope data using RKHS

11|2022 ASTRO 3D Annual Retreat, Adelaide, Australia
- ⋆ **Lead discussion:** Maximising legacy and return for Data Intensive Astronomy

08|2022 International Astronomical Union General Assembly 2022, Busan, Republic of Korea
- ⋆ **e-Talk:** Making Unexpected Time-domain Discoveries in Astronomy with Machine Learning

07|2021 Collaborative Conference on Computational and Data Intensive Science (C3DIS), Virtual
- ⋆ **Lightning talk:** Hunting the Unknowns in Murriyang, an Anomaly Detection Approach

06|2021 MARS 2021, Virtual
- ⋆ **Pitch:** Artificial intelligence for discovering rare and unknown objects in astronomical surveys

Selected Recent Supervising Experience

Total student supervision: 7

07|2023–10|2024 **Primary supervisor**, *Master of Science (Mathematics and Statistics)*, School of Physics, University of Melbourne

12|2022–03|2023 **Primary supervisor**, *Vacation Scholarship*, AAO & MQAAAstro, Macquarie University

12|2021–02|2022 **Secondary supervisor**, *Laby Research Scholars Program*, School of Physics, University of Melbourne

11|2021–02|2022 **Supervisor**, *Undergraduate Vacation Scholarship Program*, Space and Astronomy, CSIRO

Skills

Programming	Python
Data Science	Gradio, Matplotlib, NumPy, pandas, SciPy, Shiny, Streamlit
Generative AI	Anthropic Claude, Azure OpenAI, LangChain, LangGraph
ML	Hugging Face, PyTorch, scikit-learn
Dev Tools	Git, LaTeX
Languages	English, Malay (fluent); Chinese, Japanese (basic)

Updated: July 2026